Schedule INE410154 - Applied Machine Learning : 2026.1 - pós

Course Name - Code	Applied Machine Learning - INE410154	45 h-a	**Programa**	Computer Science (PPGCC)
Main goal	Provide students the required knowledge to solve practical problems with learning-based techniques, and also how to prepare and improve the data used in these approaches to build more efficient models.		**Professor**	Jônata Tyska
Learning Objectives	After attending this course, students should be able to: --> describe common learning-based techniques that can be used to solve many types of problems --> explain the main steps involved in data preparation and which techniques can be used to improve a given data set --> define a general machine learning pipeline, including techniques to avoid overfitting --> discuss how machine learning is being employed in different research fields to solve relevant problems --> apply machine learning techniques to scientific and/or commercial domains and applications --> read and critically analyze research papers applying machine learning methods to different fields --> design and implement experiments applying machine learning to their research areas --> write a research paper reporting their scientific findings regarding the application of machine learning methods --> present their scientific findings regarding the application of machine learning methods to an audience		**Contact**	communication via Moodle UFSC
			Meetings	Every Wednesday 10h to 12h30 Room INE313
			Evaluation (more details on the Syllabus)	Homework exercises (HW) + Seminars (SE) + Final Project (FP)

Week	Topic	Subject
Week 1	Introduction and Applications	Basic Concepts and Terminology - Types of Learning - The classical ML design flow - Applications
11/03	3 hours	
Week 2	Applied Machine Learning Final Project Discussions	- Presentation and discussion of the final project proposals
18/03	3 hours	
Week 3	Getting to know your data	- Data Types - Exploratory Data Analysis - Descriptive Statistics - Visualization Techniques
25/03	3 hours	
Week 4	Data preparation	Data Cleaning - Missing values and imputation techniques - Dealing with incorrect or incosistent values - Scaling and normalization - Cleaning for special types of data
01/04	3 hours	
Week 5	Data preparation	Feature Extraction and Engineering - numerical and categorical data - image data - text data - sequential data
08/04	3 hours	
Week 6	Data preparation	Dimensionality Reduction - Feature selection with filter and wrapper methods Data Reduction and Transformation - PCA, SVD and transforms
15/04	3 hours	
Week 7	Basic Learning Techniques and Concepts	- Linear and Logistic Regression - Gradient Descent
22/04	3 hours	
Week 8	Basic Learning Techniques and Concepts	- Introduction to Neural Networks (Perceptron, MLP and backpropagation)
29/04	3 hours	
Week 9	Basic Learning Techniques and Concepts	Decision Trees Ensemble methods --> Bagging --> Boosting --> Random Forests --> XGBoost
06/05	3 hours	
Week 10	Basic Learning Techniques and Concepts	- Data split --> Holdout --> Cross-Validation - Evaluation Metrics --- - Bias-Variance - Avoiding overfitting --> Regularization - Hyper-parameter tuning
13/05	3 hours	
Week 11	Basic Learning Techniques and Concepts	Clustering --> K-means --> Spectral Clustering --> DBScan
20/05	3 hours	
Week 12	**Applied machine learning seminars**	
27/05	3 horas	

Week 13	**Applied machine learning seminars**			
03/06	3 hours			
Weeks 14	**Final Projects**			
10/06	3 hours			
Weeks 15	**Final Projects**			
17/06	3 hours			